DO NOW — RECALL

Trees and binary search trees

Name: _____ Class: _____ Date: _____

1. In chain A→B→C, which node follows B?

2. What does LIFO mean?

3. What must be true before binary search on an array?

4. Insert X between A and B: what should new X.next point to?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

C.

2. Model answer

Last In, First Out.

3. Model answer

The array must be sorted.

4. Model answer

B, before changing A.next to X.

Vocabulary — know these terms

root — Topmost tree node.

child — Node directly below a parent.

subtree — A node and all its descendants.

binary search tree — A binary tree ordered by values, with smaller keys to the left and greater to the right.

balanced — Height kept relatively small compared with node count.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Confirm and explain

35 is smaller than 40, so insert it as the left child of 40.

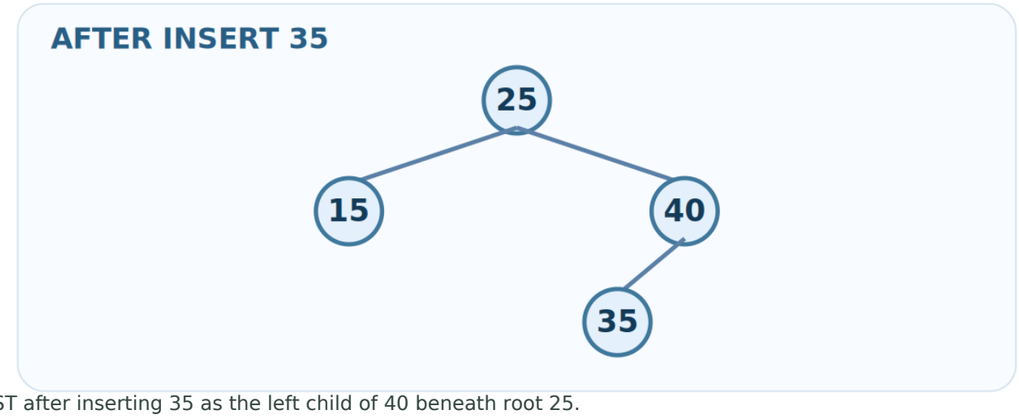
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 35 into BST with root 25, left 15, right 40.

35 is greater than 25, so go right.



BST after inserting 35 as the left child of 40 beneath root 25.

NEW LEARNING

Read, annotate and explain

Start at root

Compare the new value 25 with root 40; it belongs in the left subtree.

Compare again

25 is greater than 20, so move right. The right child currently stores 30.

Insert without breaking the property

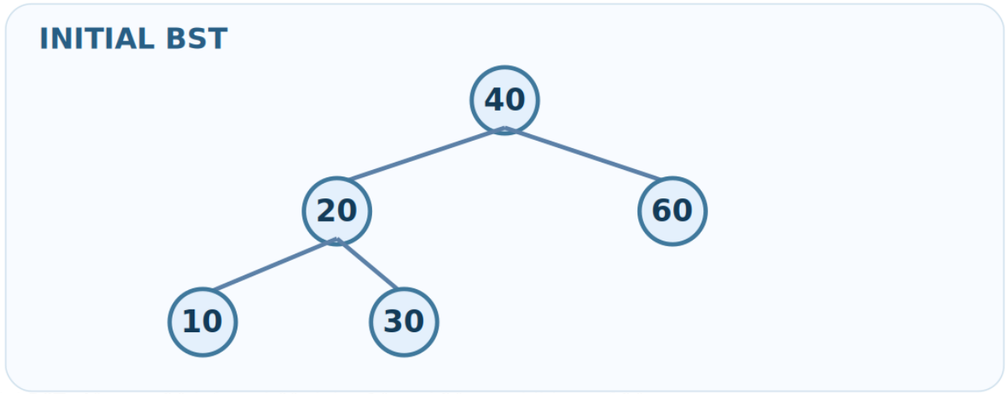
25 is smaller than 30; insert it as the left child of 30.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

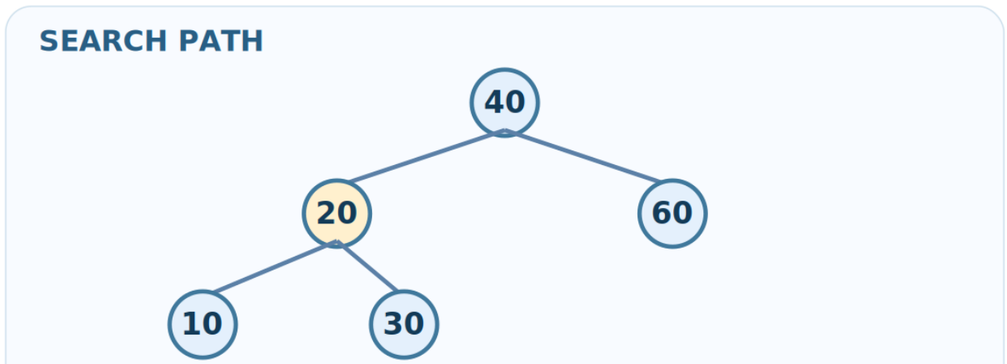
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Initial BST: 40 root; 20 left and 60 right; 10 and 30 are children of 20.

New Learning



BST search path for inserting 25: from 40 left to 20, right to 30, then left.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Every binary search tree has logarithmic worst-case lookup time.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

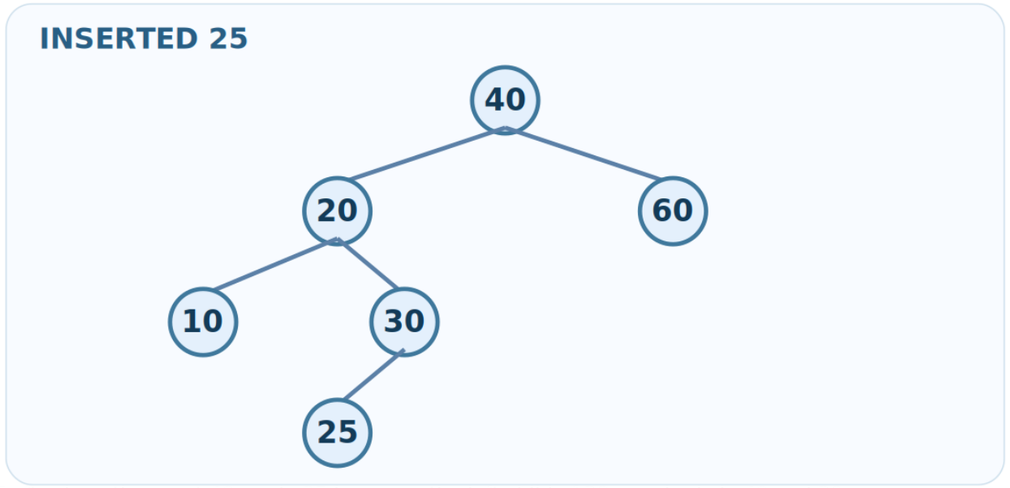
Question 3

Explain why a BST can have O(n) worst-case search time despite O(log n) average search on a reasonably balanced tree.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



BST after insertion: root 40; node 25 becomes the left child of 30, reached via 40 and 20.

I DO — WORKED EXAMPLE

Follow the worked method

Insert 12 into BST with root 20, left 10 and right 30.

Predict

Compare 12 with 20 → go left to 10.

Confirm and explain

12 > 10, so 12 becomes the right child of 10.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

BST root 50, left child 25, right child 75: where does 60 go?

Question 2

Insert 15, 10, 20, 18 into an empty BST. Identify the parent of 18.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict 35 is greater than 25, so go right.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

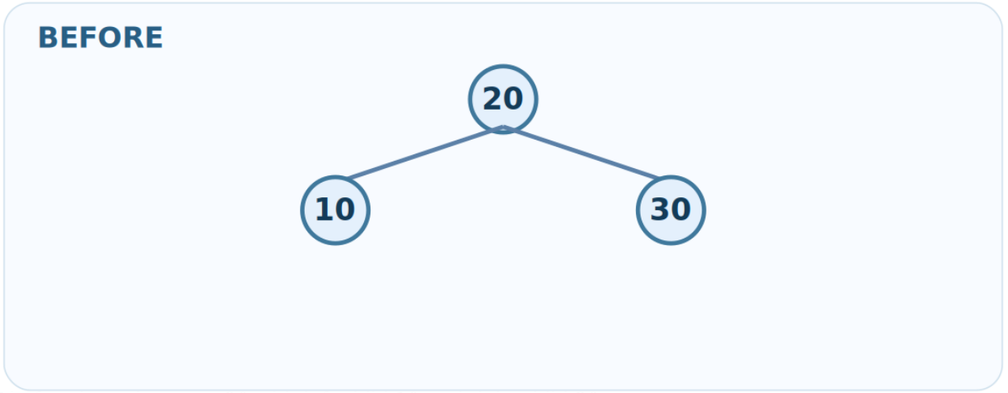
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 12 into BST with root 20, left 10 and right 30.

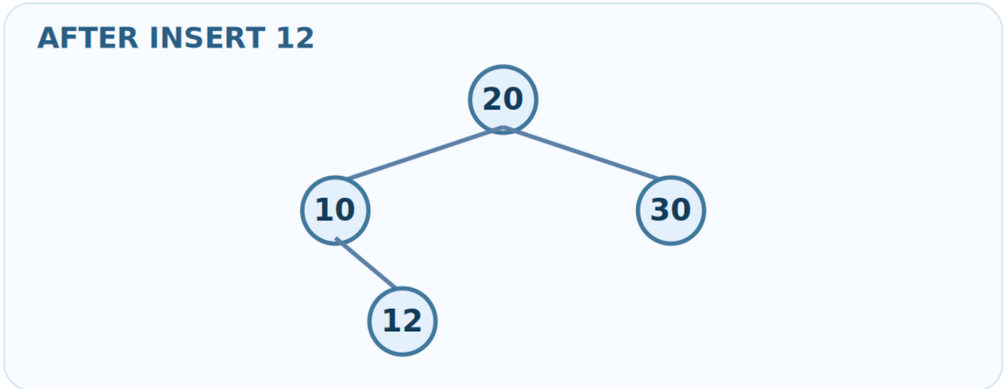
Compare 12 with 20 → go left to 10.



BST before insertion: root 20 with left child 10 and right child 30.

Insert 12 into BST with root 20, left 10 and right 30.

Compare 12 with 20 → go left to 10.



BST after inserting 12 as the right child of 10 beneath root 20.

WE DO — GUIDED PRACTICE

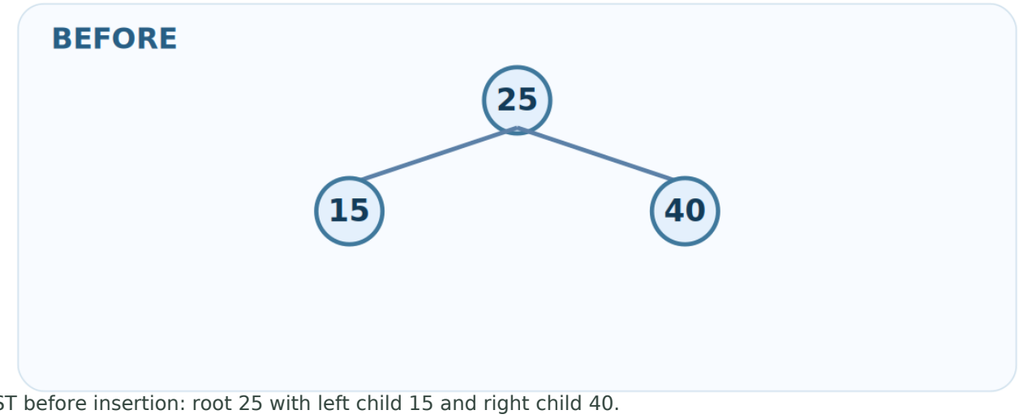
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Insert 35 into BST with root 25, left 15, right 40.

35 is greater than 25, so go right.



BST before insertion: root 25 with left child 15 and right child 40.